

IB DIPLOMA PROGRAMME · ECONOMICS

Economic Growth

Macroeconomics · Standard & Higher Level

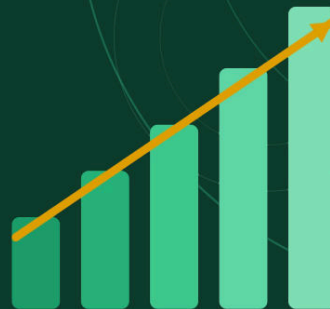
An interactive study deck — diagrams, worked examples, exercises and real exam practice, built to give you a solid understanding fast.

Diagrams

Exercises

Exam practice

► **Tip:** run this in Slide Show mode — click to reveal answers and watch the diagrams shift.



WHAT TO SAY

Welcome. Today's topic is economic growth - one of the main macroeconomic objectives. By the end you should be able to define, draw, explain, evaluate and calculate growth (SL and HL).

Tip: run this in Slide Show mode - the exercises reveal their answers on click.

HOW TO USE THIS DECK

What you will master



Define & measure growth Real GDP, real GDP per capita, and why "real" matters.

1



Show growth on diagrams The PPC and the AD-AS model (short run vs long run).

2



Explain the causes More factors of production and better productivity.

3



Evaluate the consequences Living standards, inequality, the environment & sustainability.

4



Apply & sit the exam Growth-rate calculations plus Paper 1 exam-style questions.

5

WHAT TO SAY

Quick roadmap of the five things we will master. Emphasise that diagrams and evaluation earn the most marks in the exam. Point out the 'Your turn' slides where students try a question before the answer is revealed.

SECTION

01

The Big Picture

What economic growth is — and how we measure it.



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3

WHAT TO SAY

Section 1: the big picture - what economic growth means and how we measure it.

04 What is economic growth?

01 • THE BIG PICTURE

What is economic growth?

Economic growth = an increase in the **real output (real GDP)** of an economy over a period of time.

- Measured as the % change in real GDP from one year to the next.
- Real GDP per capita (real GDP ÷ population) better reflects average living standards.
- “Real” strips out inflation, so we compare volumes of output, not just higher prices.
- It is one of the key macroeconomic objectives, alongside low unemployment, low inflation and equity.



Key terms

- Real GDP — output valued at constant (base-year) prices.
- Nominal GDP — output at current prices (inflation-distorted).
- GDP per capita — GDP ÷ population.
- Rate of growth — annual % change in real GDP.

WHAT TO SAY

Core definition: an increase in REAL GDP over time. Stress two words: 'real' (adjusted for inflation) and 'per capita' (better for living standards). Remind them growth is one of the macroeconomic objectives.

01 • THE BIG PICTURE

Real vs nominal GDP



Nominal GDP

- Measured at current (this year's) prices.
- Rises when prices rise — even if no extra goods are made.
- Can overstate real progress during inflation.



Real GDP

- Measured at constant (base-year) prices.
- Strips out inflation — shows the true change in volume.
- **The correct measure of economic growth.**



Convert: $\text{Real GDP} = (\text{Nominal GDP} \div \text{GDP deflator}) \times 100$



GO FURTHER *Increase your knowledge*

Read about the difference between the GDP deflator and the CPI, and why economists prefer chained/real measures for comparisons over time.

WHAT TO SAY

Contrast the two. Nominal GDP uses current prices, so inflation can inflate it. Real GDP uses constant prices, so it shows the true change in volume - that is what we mean by growth.

Example: if prices rise but output is flat, nominal rises but real does not.

01 • THE BIG PICTURE

Actual vs potential growth



Actual growth

- A real increase in output happening now.
- Uses up spare / idle capacity (e.g. re-employing workers).
- Often cyclical — the short-run upswing of the business cycle.
- **Closes a negative output gap.**



Potential growth

- An increase in the economy's productive capacity.
- The maximum it can sustainably produce at full employment.
- Long-run — more or better factors of production.
- **Shifts the frontier (PPC / LRAS) outward.**

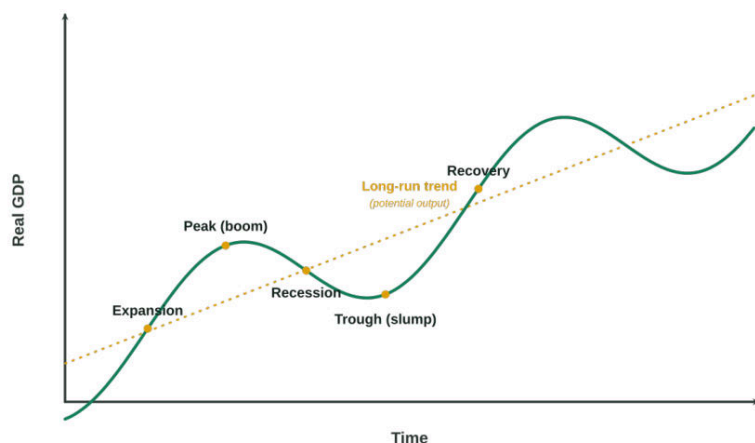
Link: closing an output gap = actual growth · expanding the capacity ceiling = potential growth.

WHAT TO SAY

The key distinction for the whole topic. Actual growth = using spare capacity now (short run, closes an output gap). Potential growth = expanding capacity (long run, shifts the frontier). We draw both on the next slides.

01 • THE BIG PICTURE

Growth & the business cycle



- **Growth is the upward long-run trend in real GDP.**
- The business cycle is the short-run fluctuation around that trend.
- Boom → recession → trough → recovery, then repeat.
- **Recession = two consecutive quarters of negative real GDP growth.**



GO FURTHER *increase your knowledge*

Explore output gaps (positive vs negative) and leading economic indicators used to forecast the cycle.

WHAT TO SAY

Growth is the long-run upward trend; the business cycle is the short-run wobble around it. Name the phases (boom, recession, trough, recovery). Define a recession: two consecutive quarters of negative real GDP growth.

SECTION

02

Show It on a Diagram

Modelling growth with the PPC and AD-AS.



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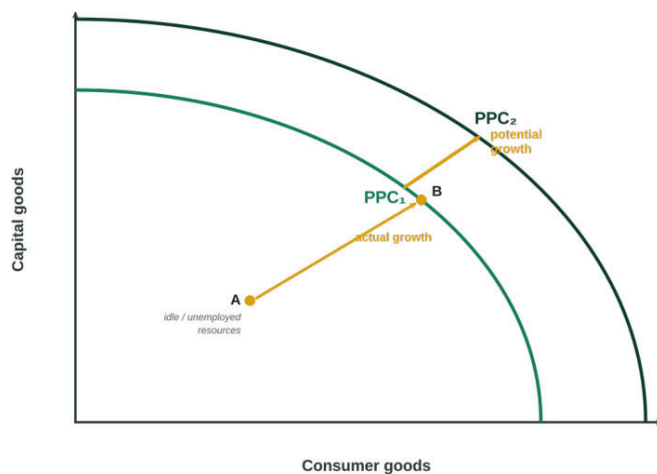
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WHAT TO SAY

Section 2: let's put growth on diagrams - the PPC and the AD-AS model.

02 • DIAGRAMS

Growth on the PPC



► Click to reveal



Actual growth

Move from a point inside the PPC (A) onto the curve (B) — using previously idle resources. The curve does not shift.



Potential growth

The whole PPC shifts outward ($PPC_1 \rightarrow PPC_2$) — more or better factors of production raise capacity.

WHAT TO SAY

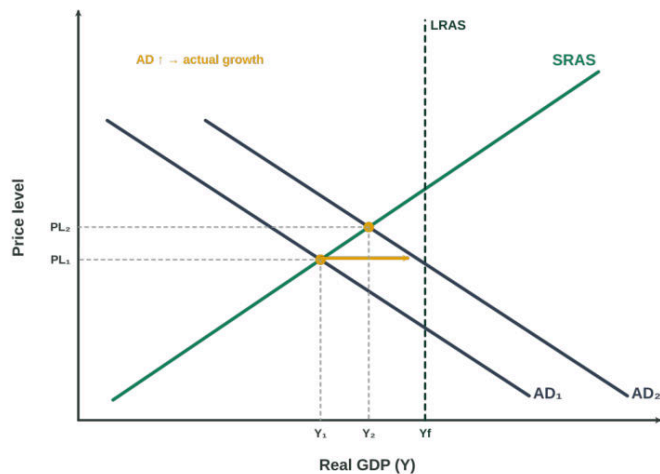
Start with the base PPC. Actual growth = moving from a point inside (A) onto the curve (B) using idle resources. CLICK to reveal potential growth = the whole curve shifting outward. One line to remember: inside-to-curve = actual; whole-curve-out = potential.

10 AD-AS: actual (short-run) growth

02 • DIAGRAMS

AD-AS: actual (short-run) growth

► Click to reveal



- A rise in aggregate demand ($AD_1 \rightarrow AD_2$) raises real GDP in the short run.
- Drivers: consumption (C), investment (I), government (G) or net exports (X-M).
- **But note: the price level also rises ($PL_1 \rightarrow PL_2$) — inflation risk as you near full capacity (Y_f).**



GO FURTHER *increase your knowledge*

This is demand-side growth. Compare it with supply-side (LRAS) growth on the next slide.

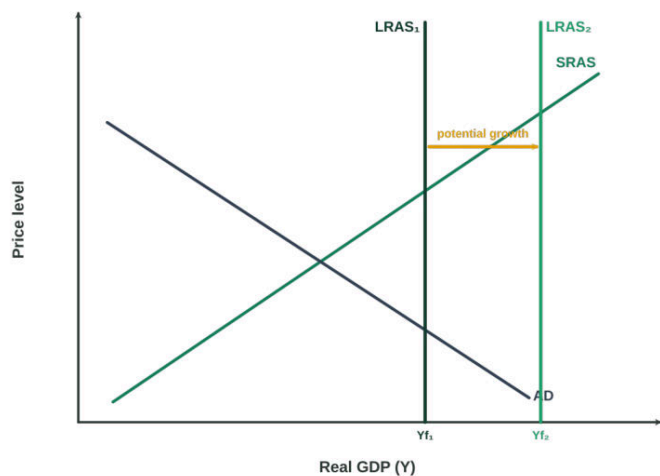
WHAT TO SAY

Show equilibrium below full capacity. CLICK to shift AD to the right - real GDP rises (actual growth), but so does the price level (inflation risk near full capacity). This is demand-side growth.

11 AD-AS: potential (long-run) growth

02 • DIAGRAMS

AD-AS: potential (long-run) growth



- Potential growth = a rightward shift of LRAS (LRAS₁ → LRAS₂).
- Caused by more / better factors of production and higher productivity.
- **Full-capacity output rises with no inflationary pressure — this is sustainable growth.**



GO FURTHER *increase your knowledge*

Compare the Monetarist/New-Classical vertical LRAS with the Keynesian AS model, where the vertical (full-employment) section shifts right.

WHAT TO SAY

CLICK to shift LRAS to the right = potential growth. Capacity rises with no inflationary pressure, so it is sustainable. For stronger students, mention the Keynesian AS version where the vertical section shifts right.

12 Your turn: read the diagram

02 • DIAGRAMS

► Click to reveal

Your turn: read the diagram



Question. A government reopens idle factories, putting unemployed workers back to work. (a) Is this actual or potential growth? (b) How is it shown on a PPC?



Model answer

(a) Actual growth.

Spare capacity is being used — the economy moves toward its existing frontier, it does not expand it.

(b) A movement from a point inside the PPC to a point on the curve.

The PPC itself does not shift (that would be potential growth).

WHAT TO SAY

Let students attempt this first. Read the question, pause, then CLICK to reveal the model answer. Reinforce: reopening idle factories = actual growth = a movement inside-to-curve, with no shift of the PPC.

SECTION

03

What Causes Growth?

The engines behind more — and better — output.



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WHAT TO SAY

Section 3: what actually causes growth? Two big buckets - more factors, and better factors.

03 • CAUSES

Cause 1 — more factors of production

Increasing the QUANTITY of resources shifts the PPC / LRAS outward.



Land

New natural resources — e.g. discovering oil, gas or mineral reserves.



Labour

A larger workforce — population growth, immigration, higher participation.



Capital

Investment in machines, factories and infrastructure.



Enterprise

More entrepreneurs combining resources and taking risks.

WHAT TO SAY

More factors of production: land, labour, capital, enterprise. Each shifts the PPC/LRAS outward. Give a quick real example for each (e.g. a new oil field, immigration).

03 • CAUSES

Cause 2 — more productive factors

Improving the QUALITY of resources raises productivity (output per worker) — the engine of long-run growth.



Human capital

Education, training and better health raise worker productivity.



Technology

Innovation and automation let the same inputs produce more.



Institutions

Property rights, rule of law and stability encourage investment.



Investment (key)

Raises both the quantity AND quality of capital — a double driver.

WHAT TO SAY

Better, more productive factors: human capital, technology, institutions, and investment (which raises both quantity and quality). Productivity - output per worker - is the real long-run engine.

16 Your turn: sort the cause

03 • CAUSES

► Click to reveal

Your turn: sort the cause

Is each a *QUANTITY* of factors or a *QUALITY* / productivity improvement? (Answers appear on click.)

1. A baby-boom generation enters the workforce

Quantity — Labour

2. A national free-vaccination programme

Quality — Human capital

3. New AI software raises output per worker

Quality — Technology

4. A large new oil field is discovered

Quantity — Land

5. A government vocational-training scheme

Quality — Human capital

6. Net inward migration of workers

Quantity — Labour

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WHAT TO SAY

Classification exercise. Read each item, let students call 'quantity' or 'quality', then CLICK to reveal. Watch-outs: training and health = quality; more workers or new land = quantity.

SECTION

04

Consequences



Benefits, costs, and the evaluation examiners love.

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17

WHAT TO SAY

Section 4: consequences - the benefits, the costs, and the evaluation examiners reward.

04 • CONSEQUENCES

Benefits of economic growth



Higher living standards

Higher average real incomes let households consume more.



Lower unemployment

Growing firms hire more workers as output expands.



More tax revenue

Funds public services and can reduce government debt.



Less absolute poverty

Rising incomes can lift people out of poverty — if shared.

Also: stronger business & consumer confidence, encouraging further investment.

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WHAT TO SAY

Quick run-through of the benefits: higher incomes and living standards, lower unemployment, more tax revenue, less poverty. Students usually know these, so keep it brisk.

04 • CONSEQUENCES

Costs & risks of growth



Demand-pull inflation

If AD-driven near capacity, the price level rises.



Environmental damage

Pollution, carbon emissions and depletion of resources.



Rising inequality

The gains may go mostly to owners of capital and skilled workers.



May be unsustainable

Fast growth today can undermine the resources of tomorrow.

Also: structural unemployment as "creative destruction" makes some industries obsolete.

WHAT TO SAY

The flip side: demand-pull inflation, environmental damage, rising inequality, and sustainability concerns. This sets up the evaluation on the next slide.

04 • CONSEQUENCES

Evaluation: is growth always good?

Top marks come from judgement — “it depends on...”



How it is achieved

AD-led growth risks inflation; LRAS-led (supply-side) growth is more sustainable.



Who gains

If growth mainly benefits the top, inequality can worsen despite a bigger economy.



Environmental cost

“Dirty” growth harms welfare; green growth aims to decouple output from emissions.



Context & type

A little growth means more in a low-income country than in a rich one.

Exam tip: always weigh benefits against costs, use a real-world example, then give a supported final judgement.

WHAT TO SAY

The single most important exam skill. Growth is not automatically good - it depends on how it is achieved, who gains, the environmental cost, and the country's context. Drill the exam tip: weigh both sides, use an example, give a judgement.

04 • CONSEQUENCES

Growth vs development



Economic growth

Quantitative: simply more real output (GDP). A necessary means, but narrow — it says nothing about how people actually live.



Economic development

Multidimensional: incomes + health + education + freedoms & choices. Growth can enable it, but is not sufficient on its own.



GO FURTHER *increase your knowledge*

Read about: GDP vs the HDI · sustainable development (meeting today's needs without harming future generations) · “green growth” and decoupling growth from carbon emissions.

WHAT TO SAY

Growth = more output (quantitative). Development = broader wellbeing - incomes plus health, education and freedoms. Growth can enable development but is not the same thing.

SECTION

05

Crunch the Numbers

The quantitative skills the exam rewards.



WHAT TO SAY

Section 5: the numbers - the calculation skills the exam expects.

05 • QUANTITATIVE SKILLS

Calculating economic growth



Rate of economic growth

The annual percentage change in real GDP.

$$\% \Delta = [(GDP_2 - GDP_1) \div GDP_1] \times 100$$



Real GDP from nominal

Removes the effect of inflation.

$$\text{Real GDP} = (\text{Nominal GDP} \div \text{deflator}) \times 100$$



Real GDP per capita

Best proxy for average living standards.

$$= \text{Real GDP} \div \text{Population}$$

WHAT TO SAY

Three formulas. The growth-rate formula is the one most tested. Show the deflator conversion and the per-capita formula quickly.

24 Worked example

05 • QUANTITATIVE SKILLS

► Click to reveal

Worked example



Problem. Real GDP is \$500bn in Year 1 and \$530bn in Year 2. (a) Find the growth rate. (b) If population rose 2%, what happened to real GDP per capita?

(a) Growth rate

$$\begin{aligned} &= (530 - 500) \div 500 \times 100 \\ &= 30 \div 500 \times 100 \\ &= \mathbf{6\%} \end{aligned}$$

(b) Real GDP per capita

Approx. $\% \Delta$ per capita $\approx \% \Delta$ GDP $- \% \Delta$ population

$$\approx \mathbf{6\% - 2\% = +4\%}$$

Living standards rose — output grew faster than population.

Tip: the subtraction rule is an approximation — fine for small percentages.

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WHAT TO SAY

Do part (a) live: $30/500 \times 100 = 6\%$. CLICK to reveal. Then part (b): per-capita growth is about $6\% - 2\% = 4\%$. Emphasise that the subtraction shortcut works for small percentages, and that living standards rose.

SECTION

06

Exam Zone

Quick-fire quiz and Paper 1 practice.



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WHAT TO SAY

Section 6: exam zone - let's test it.

Quick-fire quiz

▶ Click to reveal

1. Economic growth is best defined as an increase in...

A) nominal GDP B) real GDP over time C) the money supply

✓ B) real GDP over time

2. An outward shift of the PPC represents...

A) actual growth B) inflation C) potential (long-run) growth

✓ C) potential (long-run) growth

3. Which is most likely a cost of rapid growth?

A) lower tax revenue B) environmental degradation C) lower prices

✓ B) environmental degradation

WHAT TO SAY

Ask each question, let students answer, then CLICK to reveal the correct option. A quick, fun recap of the key definitions and the PPC point.

06 • EXAM ZONE

Paper 1 — exam-style questions

10 marks

Part (a)

“Explain how investment can increase an economy’s rate of economic growth.”

- Define economic growth & investment.
- Use a diagram: LRAS (or PPC) shifting right.
- Explain the chain: investment → more/better capital → productivity → capacity.
- One real-world example.

15 marks

Part (b)

“Evaluate the view that economic growth always improves living standards.”

- Benefits: incomes, jobs, tax revenue, poverty.
- Against: inequality, inflation, environment, sustainability.
- “Depends on” how growth is achieved & shared.
- **Real-world example + a supported judgement.**

Command terms: “Explain” = give reasons with theory + diagram. “Evaluate” = weigh both sides and judge.

WHAT TO SAY

Show how to structure Paper 1 answers. (a) 'Explain' = theory + a diagram + a chain of reasoning + a real example. (b) 'Evaluate' = both sides + a supported judgement. Point out the command terms.

RECAP

Key terms glossary

Economic growth — increase in real GDP over time.

Real GDP — output at constant (base-year) prices.

Nominal GDP — output at current prices.

GDP per capita — $\text{GDP} \div \text{population}$.

Actual growth — using spare capacity (inside → onto PPC).

Potential growth — rise in capacity (PPC / LRAS shifts out).

PPC — max combinations of goods with full resources.

Aggregate demand — $C + I + G + (X - M)$.

LRAS — output at full employment (potential).

Productivity — output per unit of input (per worker).

Recession — two quarters of negative real growth.

Sustainable growth — without harming future generations.

WHAT TO SAY

A recap of every key term - great for revision. Tell students to cover the right-hand column and test themselves on the definitions.

WRAP-UP

Key takeaways & where to go next



Take these away

- Growth = rising real GDP; per-capita real GDP tracks living standards.
- Actual growth uses spare capacity; potential growth expands it.
- Show it with the PPC (shift out) and AD-AS (AD vs LRAS).
- Causes = more factors + higher productivity (investment & tech are key).
- **Always evaluate: growth has real costs, so it is not automatically “good”.**

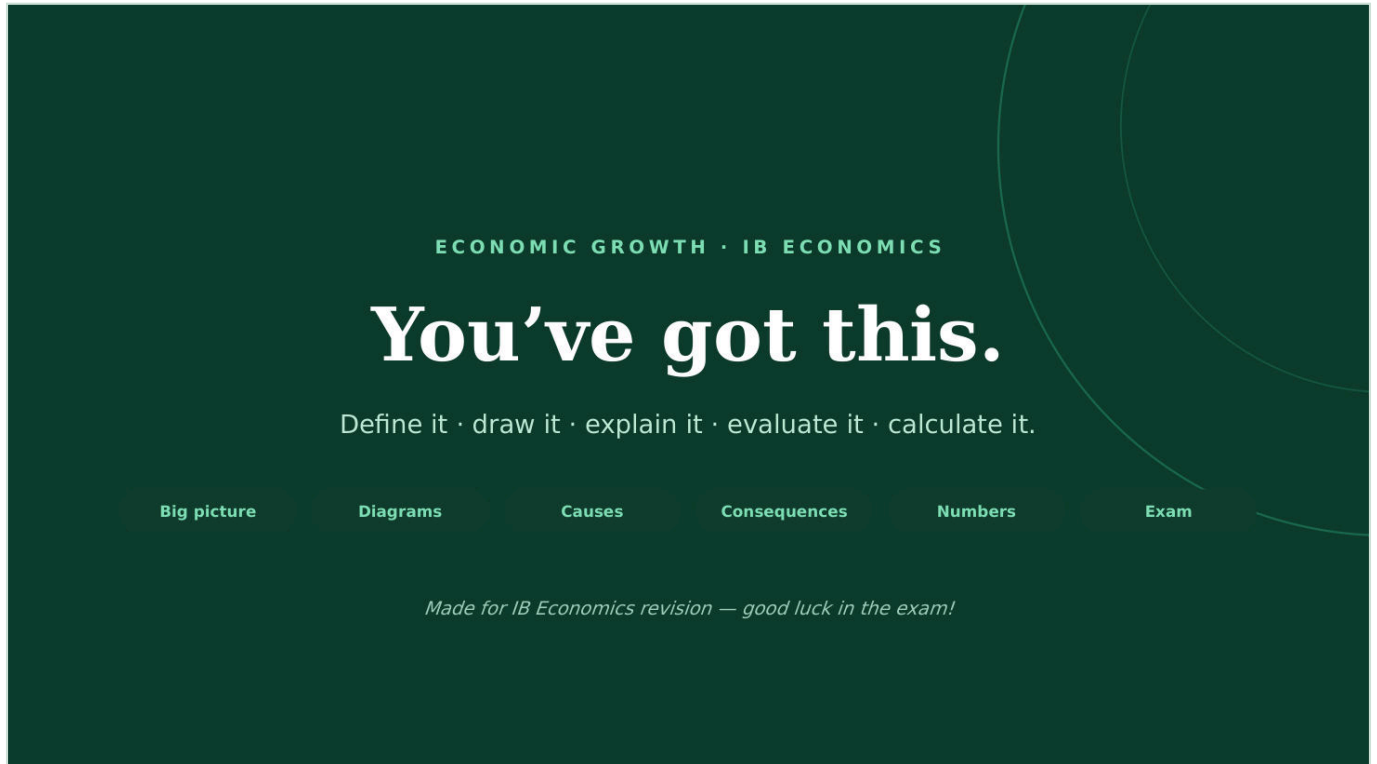


Go further

- Output gaps & the Keynesian multiplier (HL).
- Monetarist vs Keynesian AS models.
- Endogenous vs exogenous growth theory.
- GDP vs HDI — growth vs development.
- Green growth, decoupling & the Kuznets curve.
- A real data-response on one country's growth.

WHAT TO SAY

Five takeaways - especially: always evaluate. Point to the 'go further' list for anyone who wants to push beyond the syllabus (HL multiplier, endogenous growth, HDI, green growth).



WHAT TO SAY

Wrap up with the five verbs: define it, draw it, explain it, evaluate it, calculate it. Invite questions. Good luck in the exam!